Lecture 12 & 13

2/2/2016.
– 4/2/2016.

Complete Integrals and General Solutions

We considered general solutions for quasilinear equations by the method of Lagrenge, which produced a solution depending on an arbitrary function.

Now the question is do such general solutions exist for fully nonlinear equations? The answer is YES, but the process is more complicated than in the quasilinear case.

Let us first study some simple classes of solutions called complete Integrals.

A complete integral for $F(x,y,z,p,q)=0$ is a solution $u(x,y;a,b)$ that depends on independent parameters a and b. The independence of the parameters means that the mapping $(a, b) \longmapsto (u, u_x, u_y)$ should have rank 2 at each fixed x, y. (2)

<u>For example</u>: Revisit the PDE: $u_x u_y = u$

Suppose $u(x,y) = X(x)\, Y(y)$. (for $X(x) \not\equiv 0$, $Y(y) \not\equiv 0$)

Then $u_x = X'(x) \cdot Y(y)$

$u_y = X(x)\, Y'(y)$.

$$u_x u_y = u \Rightarrow X'(x)\, Y(y) \cdot X(x)\, Y'(y) = X(x) Y(y)$$

$$\Rightarrow X'(x)\, Y'(y) = 1$$

$$\Rightarrow X'(x) = \frac{1}{Y'(y)} = \mu \text{ (constant)}$$

$$X'(x) = \mu \Rightarrow X(x) = \mu x + c_1$$

$$Y'(y) = \frac{1}{\mu} \Rightarrow Y(y) = \frac{1}{\mu} y + c_2.$$

$$u(x,y) = X(x) \cdot Y(y) = (\mu x + c_1)\left(\frac{1}{\mu} y + c_2\right)$$

$$= xy + c_2 \mu x + \frac{c_1 y}{\mu} + c_1 c_2.$$

or $xy + ax + by + ab$.

So $u_x u_y = u$ admits solutions of the form $u(x,y) = xy + ax + by + ab$

Consider the map $(a,b) \longmapsto (u_x, u_y)$.

$$D \equiv \begin{vmatrix} u_{xa} & u_{ya} \\ u_{xb} & u_{yb} \end{vmatrix}$$

For the example: $u_x = y + a$,

$u_{xa} = 1, \quad u_{xb} = 0$

$u_y = x + b$

$u_{ya} = 0, \quad u_{yb} = 1$

$$D = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0.$$

This condition $D \neq 0$ for the independence of the parameters a, b is only sufficient condition but not necessary condition.

Example: Here is a PDE $u_x^2 + u_y = 0$ whose solution has the form

$$u(x, y; a, b) = ax - ya^2 + b$$

but $D \equiv 0$. (Because $u_{xa} = 1$

$u_{xb} = 0$

$$D = \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$u_{ya} = -2a$

$u_{yb} = 0.$)

How to obtain the solution?

$$\frac{dx}{F_p} = \frac{dy}{F_q} = \frac{dz}{pF_p + qF_q}$$

$$= \frac{dp}{0} = \frac{dq}{0}$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$q = b \text{ (const)}$$

$$p = a \text{ (Constant)}$$

$$dz = a\,dx + q\,b\,dy$$

$$\Rightarrow z = ax + by + c$$

ie $u(x,y) = ax + by + c$

$$u_x = a, \quad u_y = b$$

$$u_x^2 + u_y = 0 \Rightarrow a^2 + b = 0$$

$$\Rightarrow b = -a^2.$$

$$\therefore u(x,y) = ax - a^2y + c$$

or

$$u(x,y) = ax - a^2y + b.$$

The parameters are a, b.

Are there solutions other than complete integrals?
Complete integrals are 2-parameter families.
So let us look one-parameter families.

If S_a is a one-parameter family of integral surfaces for $F(x, y, z, p, q) = 0$ then so is the envelope $\mathcal{E}$.
This is obvious because every tangent plane of $\mathcal{E}$ coincides with a tangent plane of one of the S_a.
We cannot take the envelope of the Complete integral since it depends on two parameters.
But we first choose an arbitrary function 'w' relating the parameters a and b, ie $b = w(a)$, then we may consider the envelope $\mathcal{E}_w$ of the

one-parameter family of surfaces given by the graphs of $u(x, y; a, w(a))$. This is our general solution depending on "w".

<u>Example</u>: Revisit the PDE $u_x u_y = u$

$$u(x, y; a, w(a)) = xy + ax + w(a)y + aw(a).$$

To see that this expression is more general than the complete integral let us take $w(a) = a$.

Then $u(x, y; a, a) = xy + ax + ay + a^2$

$$\frac{\partial u}{\partial a}(x, y; a, a) = x + y + 2a = 0$$

$$\Rightarrow a = -(x+y)/2.$$

Substituting this into $u(x, y; a, a)$

$$= xy - \frac{(x+y)}{2}x - \frac{(x+y)}{2}y + \frac{(x+y)^2}{4}$$

$$= xy - \frac{x^2}{2} - \frac{y^2}{2} - xy + \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{2}$$

$$= -(x-y)^2/4.$$

This solution $u(x,y) = -\dfrac{(x-y)^2}{4}$ is not included in the complete integral $u(x,y) = xy + ax + by + ab$.

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Application to Geometrical optics

In geometrical optics, wave propagation is studied only in terms of the position of the wave front.

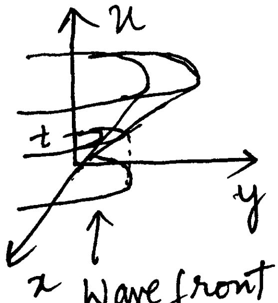
If we have a light wave in the xy-plane then its wave front at any time t is a level curve: $u(x,y) = t$.

If $c(x,y)$ denotes the propagation speed of a wave at (x,y), then the curve $(x(t), y(t))$ is a light ray if at each point its speed coincides with $c(x,y)$ and its direction is orthogonal to the wave front. (8)

Speed $c(x,y) = |(\dot{x}(t), \dot{y}(t))|$

(modulus of the velocity vector.

or



$$\sqrt{(\dot{x})^2 + (\dot{y})^2} = c(x,y).$$

Direction vector is ∇u

is orthogonal to the wave front $(x(t), y(t))$.

ie ∇u is parallel to the $(\dot{x}(t), \dot{y}(t))$.

Differentiating $u(x(t), y(t)) = t$ we get

$$u_x \dot{x} + u_y \dot{y} = 1$$

or

$$(u_x, u_y) \cdot (\dot{x}, \dot{y}) = 1$$

$$\nabla u \cdot (\dot{x}(t), \dot{y}(t)) = 1$$

But ∇u and $(\dot{x}(t), \dot{y}(t))$ are parallel;

$\Rightarrow |\nabla u| \cdot |\nabla u| = 1$ or $\|\nabla u\|^2 = 1$.

So angel between them is zero.

$$\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}| \cos\theta.$$
$$= |\vec{a}|\,|\vec{b}|.$$

Here $(\nabla u) \cdot (\dot{x}(t), \dot{y}(t))$

$$= |\nabla u| \; |(\dot{x}(t), \dot{y}(t))|$$
$$= \sqrt{u_x^2 + u_y^2} \cdot \sqrt{(\dot{x})^2 + (\dot{y})^2}$$
$$= 1.$$
$$\Rightarrow \quad c^2 (u_x^2 + u_y^2) = 1.$$

This equation is called the eikonal equation of geometrical optics in 2 dimensions.

Let us solve this PDE:

$$F(x, y, z, p, q) \equiv \frac{1}{2}[c^2 (p^2 + q^2) - 1]$$

The characteristic equations are:

$$\frac{dx}{dt} = F_p = c^2 p$$

$$\frac{dy}{dt} = F_q = c^2 q$$

$$\frac{dz}{dt} = p\,c^2 p + q\,c^2 q$$
$$= c^2(p^2+q^2) = 1.$$

$$\frac{dp}{dt} = -F_x - F_z \cdot p = 0 \Rightarrow p \text{ is const}$$

$$\frac{dq}{dt} = -F_y - F_z \cdot q = 0 \Rightarrow q \text{ is const.}$$

For initial values $(x_0, y_0, z_0, p_0, q_0)$
with $c^2(p_0^2 + q_0^2) = 1$

$$\frac{dx}{dt} = c^2 p_0 \Rightarrow x = c^2 p_0 t + x_0$$

$$\frac{dy}{dt} = c^2 q_0 \Rightarrow y = c^2 q_0 t + y_0.$$

$$\frac{dz}{dt} = 1 \Rightarrow z = t + z_0.$$

The characteristic curve is the straight line

$$x = c^2 p_0 t + x_0$$
$$y = c^2 p_0 t + y_0$$
$$z = t + z_0$$

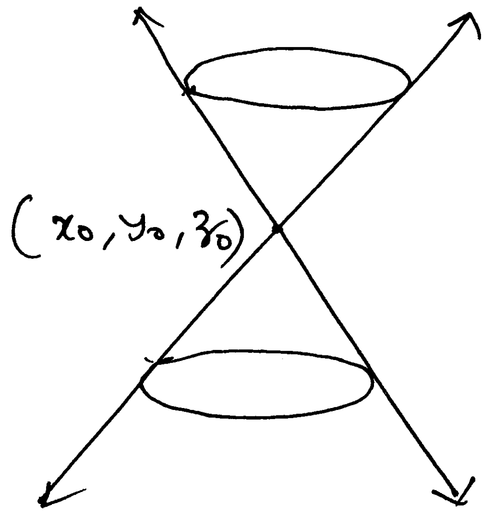
which lies on the Monge cone.

It is therefore natural to identify 't' as time (elapsed from some intial time z_0).

Moreover $(x(t), y(t))$ satisfies the conditions $\sqrt{[\dot{x}(t)]^2 + [\dot{y}(t)]^2} = c(x,y)$

$(\dot{x}, \dot{y})$ and ∇u are parallel.

for a light ray, so we identify the characteristics of $c^2(u_x^2 + u_y^2) = 1$ with light rays. Since the Monge cone is just the union of all light rays passing through the point (x_0, y_0, z_0), we call it the light cone.



The light cone.

For the initial value problem let us suppose Γ is parametrized as

$$x = f(s), \quad y = g(s), \quad z = h(s).$$

We complete Γ to a strip by choosing ϕ and ψ to satisfy

$$(1) \leftarrow c^2 [\phi^2(s) + \psi^2(s)] = 1$$

$$(2) \leftarrow h'(s) = \phi(s) f'(s) + \psi(s) g'(s).$$

From 2 $\phi = (h' - \psi g')/f'$

when $f' \neq 0$.

Substituting this ϕ in 1.

$$c^2 \left[\frac{h'^2 + \psi^2 g'^2 - 2h'g'\psi}{f'^2} \right]$$

$$+ c^2\psi^2 = 1.$$

$$\Rightarrow \overbrace{[c^2 g'^2 + c^2 f'^2]}^{A}\psi^2 +$$

$$\underbrace{(-2h'g'c^2)}_{B}\psi + \underbrace{(c^2 h'^2 - f'^2)}_{C} = 0$$

This is quadratic in ψ, so there are two solutions of ψ when $B^2 - 4ac > 0$ ie

$$4h'^2 g'^2 c^4 - 4(c^2 g'^2 + c^2 f'^2)$$
$$(c^2 h'^2 - f'^2) > 0$$

ie. $c^2 g'^2 f'^2 - c^4 f'^2 h'^2 + c^2 f'^4 > 0$

or $\quad c^2 h'^2 < g'^2 + f'^2.$

And we cannot solve the quadratic equation when $B^2 - 4AC < 0$ ie

$$c^2 h'^2 > g'^2 + f'^2.$$

So in this case we say Γ is timelike and the IVP cannot be solved.

When $c^2 h'^2 < g'^2 + f'^2$ holds we say Γ is spacelike and we can solve the IVP with two different solutions.

Now let us suppose Γ lies in the xy-plane, so $h(s) = 0$ and Γ is spacelike. Once we determine ϕ and ψ, we can use them as initial conditions in our solution of the characteristic equations:

$$\frac{dx}{dt} = c^2 p \,, \quad \frac{dy}{dt} = c^2 q$$

$$\frac{dz}{dt} = c^2 p^2 + c^2 q^2 = 1 .$$

$$\frac{dp}{dt} = 0 , \quad \frac{dq}{dt} = 0 .$$

to find

$$x(s,t) = c^2 \phi(s)\, t + f(s) .$$
$$y(s,t) = c^2 \psi(s)\, t + g(s)$$
$$z(s,t) = t \quad \text{(note that } h(s) = 0)$$

$$p(s,t) = \phi(s)$$
$$q(s,t) = \psi(s).$$

If we can solve for s,t in terms of x, y, then the above will provide the desired solution.

<u>Example</u>: PDE: $c^2(u_x^2 + u_y^2) = 1$
with $\Gamma : (\cos s, \sin s, 0)$.

To complete Γ to a strip, we must find ϕ and ψ satisfying
$c^2[\phi^2 + \psi^2] = 1$ and

$$\phi(s)(-\sin s) + \psi(s)(\cos s) = 0.$$
$$\Downarrow$$
$$\psi(s) = (\tan s)\,\phi(s).$$

First equation $\Rightarrow [c^2(\phi^2 + \tan^2 s\, \phi^2)] = 1$

$$\Rightarrow c^2(1 + \tan^2 s)\,\phi^2 = 1.$$

$$\Rightarrow \phi = \pm \frac{1}{c} \cos s.$$

and $\psi(s) = \pm \frac{1}{c} \sin s.$

$$x(s,t) = c^2 \left\{ \pm \frac{\cos s}{c} \right\} t + \cos s.$$

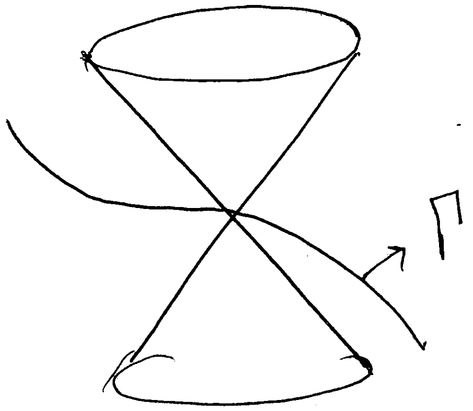
$$= \cos s \{ 1 \pm ct \}$$

$$y(s,t) = \sin s \{ 1 \pm ct \}.$$

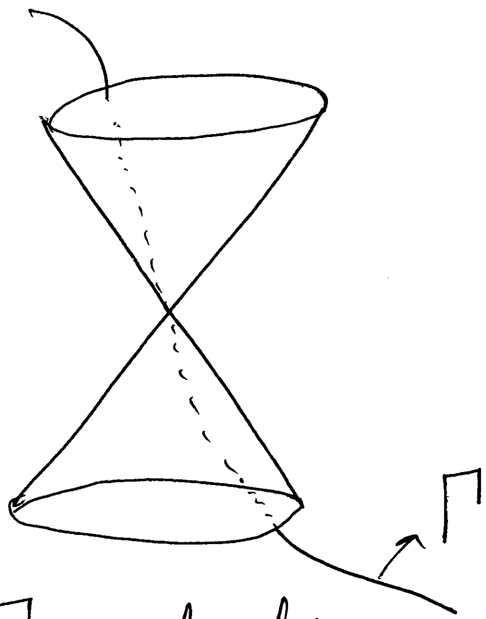
In other words we can write

$$x^2 + y^2 = (1 \pm ct)^2. \longrightarrow (*)$$

These two solutions of the IVP correspond to two distinct physical situations. Taking the +ve sign in (*) produces a wavefront moving away from the origin and taking the negative sign produces a wave front which will converge on the origin at time $1/c$.

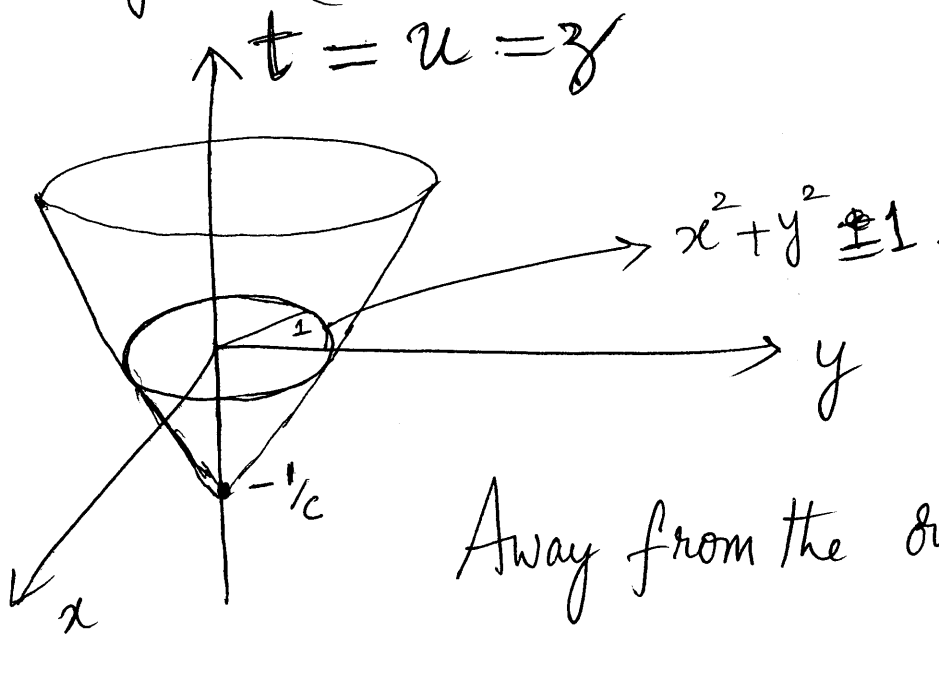


Γ spacelike

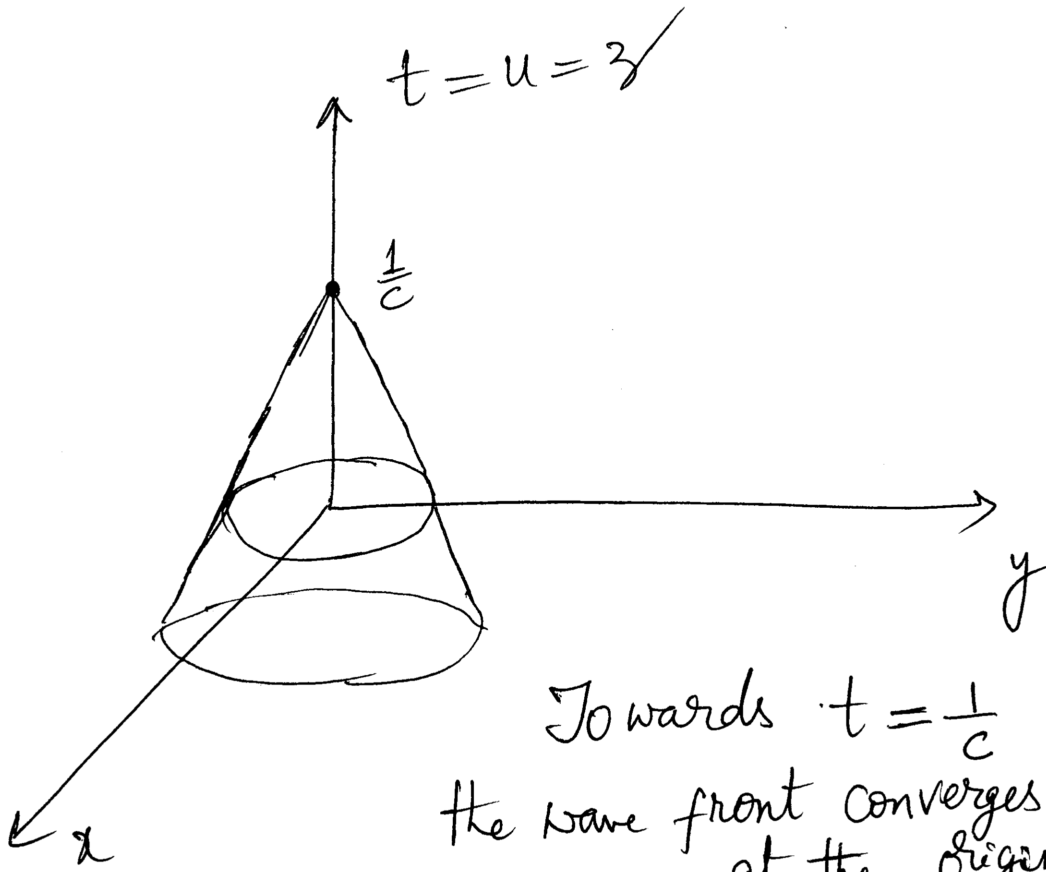


Γ is timelike.

$$x^2+y^2=(1+ct)^2.$$



Away from the origin



Towards $t = \frac{1}{c}$ the wave front converges at the origin.

$$x^2 + y^2 = (1 - ct)^2.$$

We may also study the solutions of eikonal equation of geometric optics by means of the complete integral and envelops.

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